Exercises 1-4 for Basic Concepts and the Laws of Gases

1.0.1 Describe an experimental method, based on the ideal gas law, to obtain the molecular mass of a gas.

Recall the molar form of the ideal gas law: $pV = nRT$. We know that $n = m/M$ where m is the mass of the molecule and M is the molar mass, so we have $pV = \frac{m}{M}RT$. We also know that the molar mass is constant for a given material.

An experiment which would find the molecular mass of a gas would isolate changes in m. We may also recall that the ideal gas constant $R = k_B \cdot N_A$ so that we have: $pV = \frac{m}{M}(k_B T)N_A$. Grouping together k_B and T we get the thermodynamic measure of temperature so: $pV = \frac{m}{M}\mathcal{T}N_A$. We can also notice that $M/N_A = m_\mu$ where m_μ is the microscopic mass of the molecule we get: $pV = \frac{m}{m_\mu}\mathcal{T}$ which simplifies to be:

$$m_\mu = \frac{m\mathcal{T}}{pV}$$

We know that everything in the ideal gas law cannot be kept constant without violating the equality, so we run two experiments;
Experiment number 1, allows for a change in mass and fixes temperature and pressure
Experiment number 2, allows for a change in mass and fixes temperature and volume
We can reconcile the results of those two experiments to calculate the molecular mass.

1.0.2 The density of dry air at $p = 1.0$ bar and $T = 300K$ is $1.161 kg \cdot m^{-3}$. Assuming that it consists entirely of N_2 and O_2 and using the ideal gas law, determine the amount of each gas in moles in a volume of $1m^3$ and their mole fractions.

For the ideal gas law we have $pV = n_{total}RT$ and we are able to solve for n_{total} to get $n_{total} = \frac{pV}{RT} = \frac{1.0\times10^5 Pa\cdot 1m^3}{8.314 J\cdot mol^{-1}\times 300K} \approx 40.091 mol$
Now that we have the total number of moles we can calculate the gas of the total mixture: $\rho \times V = m_{mixture} = 1.161 kg \cdot m^{-3} \cdot 1m^3 = 1.161kg = 1161g$

We know that $n_{O_2} + n_{N_2} \approx 40.091$ and that $n_{N_2}(28.014) + n_{O_2}(31.999) = 1161$.
We take $n_{O_2} \approx 40.091 - n_{N_2}$ and say that: $n_{N_2}(28.014) + (40.091 - n_{N_2})(31.999) = 1161$ Solving for n_{N_2} we get around 30.59 mol.
We now substitute this back into the earlier expression to obtain mols of O2, giving 9.50 mol.
We express our molar fractions as 9.50/40.091 and 30.59/40.091

1.0.3 The molecular density of interstellar gas clouds is about 10^4 molecules/mL. The temperature is approximately 10K. Calculate the pressure. (The lowest vacuum obtainable in the laboratory is about three orders of magnitude larger).

> We take the molecular version of the ideal gas law and obtain our answer:
> $P = (\frac{N}{V})k_B T = (10^{10} molecules/m^3)(1.38 \times 10^{-23})(10)$

1.0.4 A sperm whale dives to a depth of more than 1.5 k, into the ocean to feed. Estimate the pressure the sperm whale must withstand at this depth. (express your answer in atmospheres.)

> We can say that the total pressure is equal to the surface pressure plus the water pressure. We can also say that the water pressure increases by 1 atm every 10 meters. We take $1500/10 = 150$ and add 1 assuming standard pressure at the surface to get 151 atm.